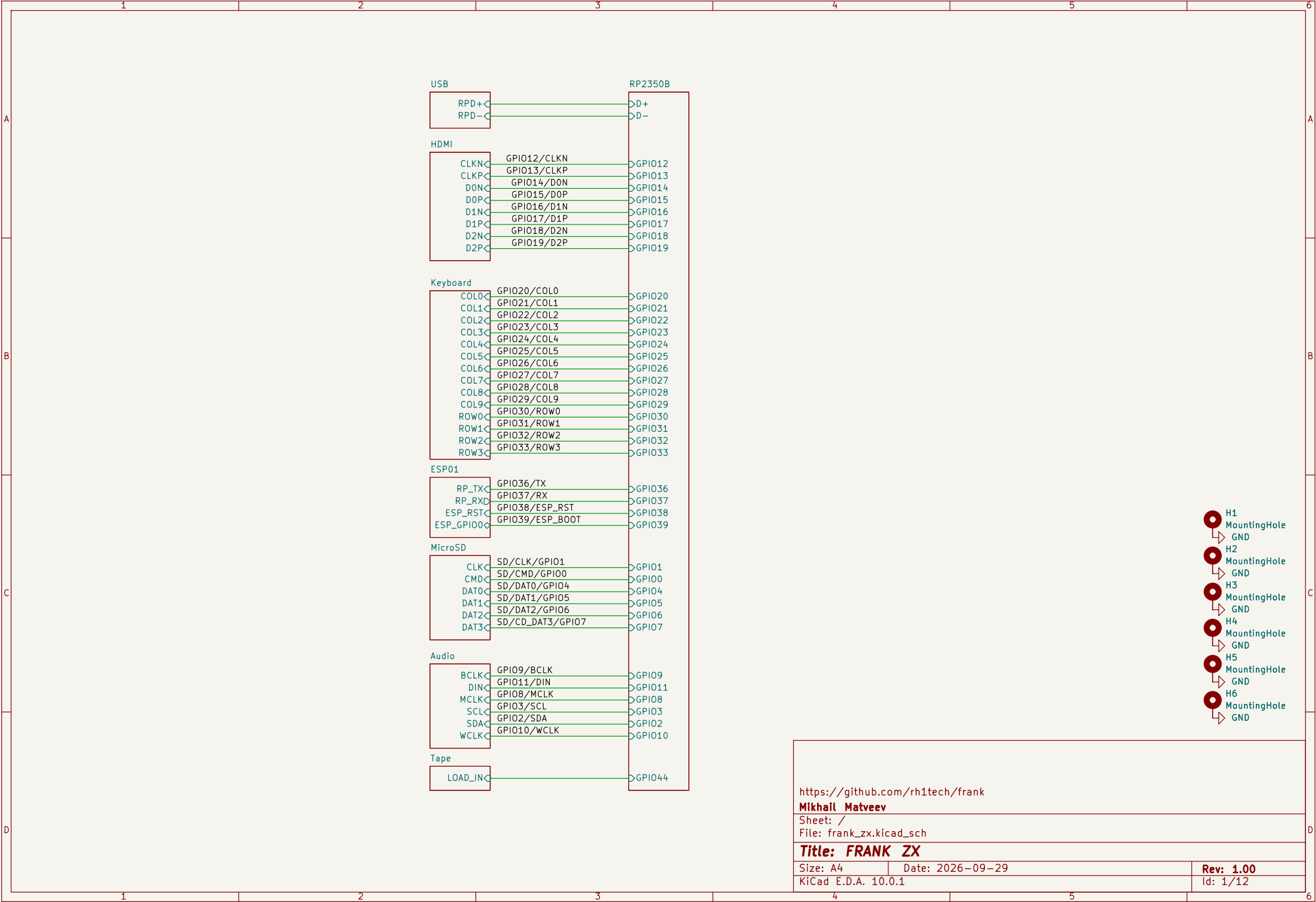
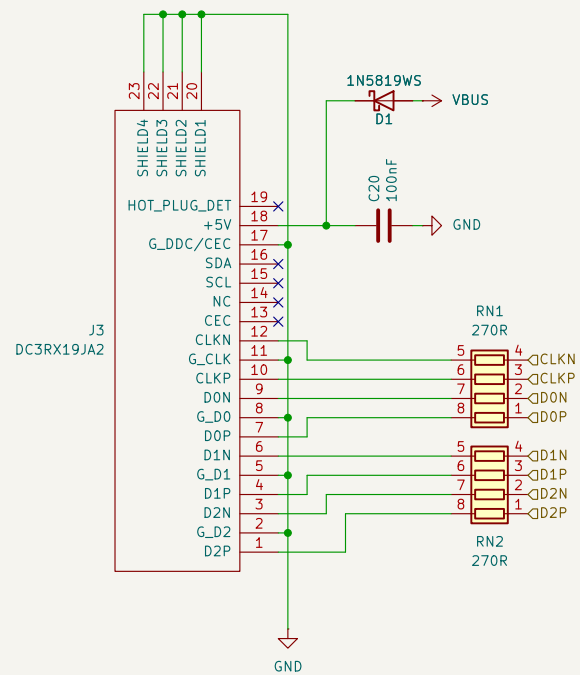






<https://github.com/rh1tech/frank>

Mikhail Matveev

Sheet: /HDMI/

File: hdmi.kicad_sch

Title: FRANK ZX

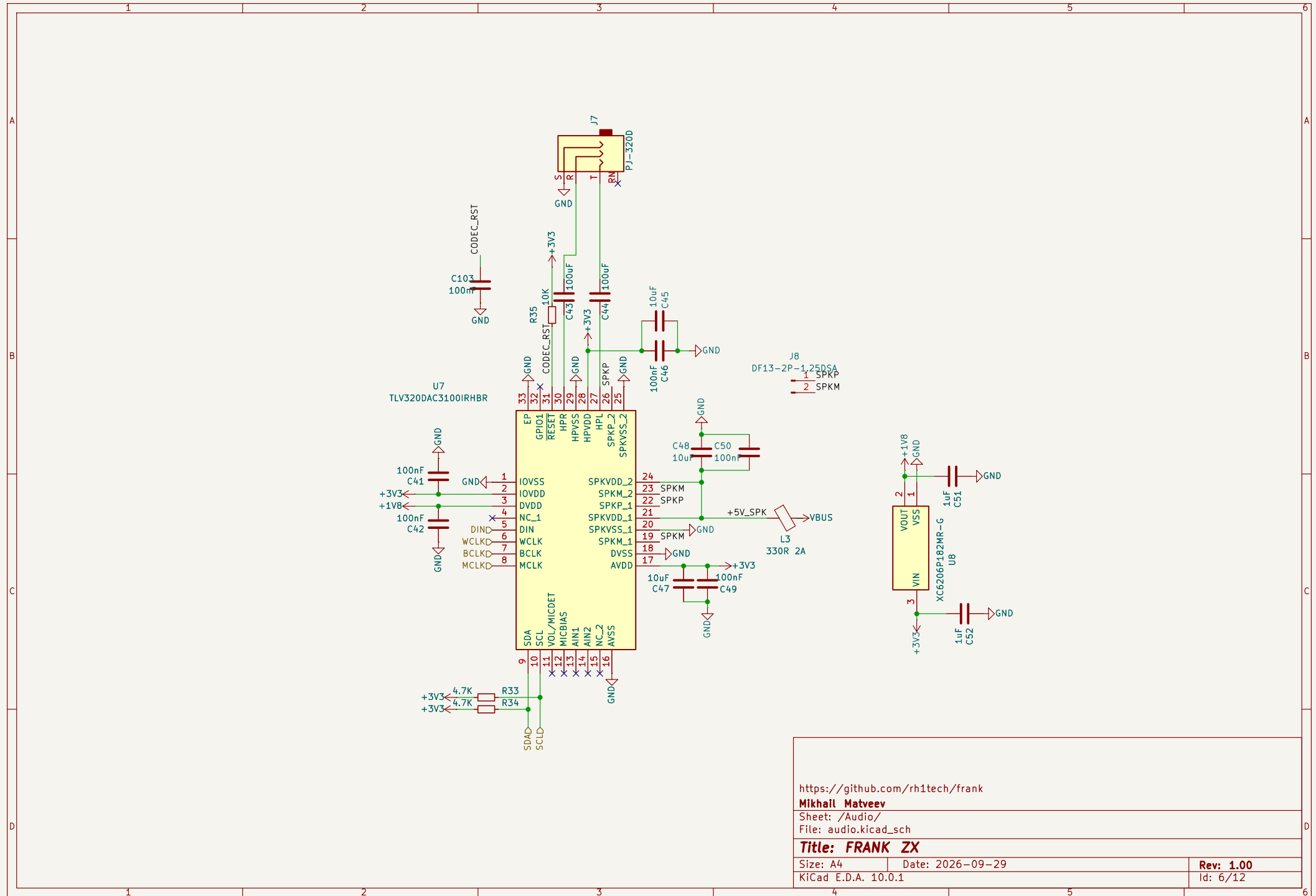
Size: A4 Date: 2026-09-29

KiCad E.D.A. 10.0.1

Rev: 1.00

Id: 3/12

Id: 5/12



<https://github.com/rh1tech/frank>

Mikhail Matveev

Sheet: /Audio/

File: audio.kicad_sch

Title: FRANK ZX

Size: A4

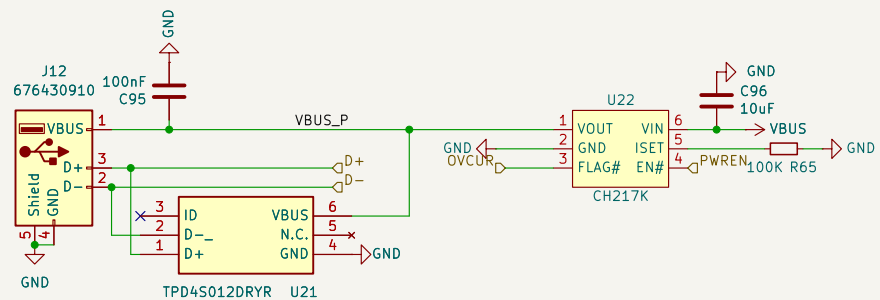
Date: 2026-09-29

KiCad E.D.A. 10.0.1

Rev: 1.00

Id: 6/12

Id: 8/12



<https://github.com/rh1tech/frank>

Mikhail Matveev

Sheet: /USB/USB A Port 1/

File: usb_a_port.kicad_sch

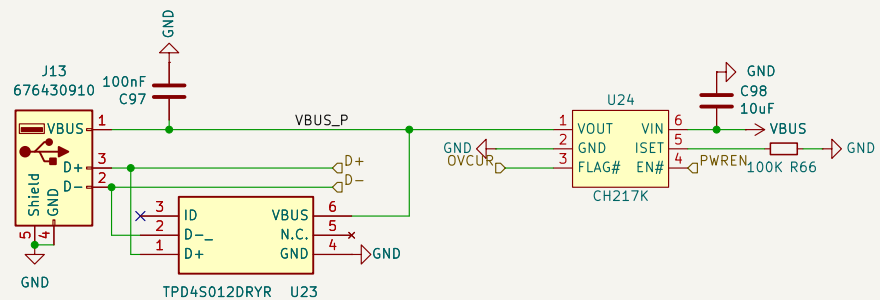
Title: FRANK ZX

Size: A4 Date: 2026-09-29

KiCad E.D.A. 10.0.1

Rev: 1.00

Id: 12/12



<https://github.com/rh1tech/frank>

Mikhail Matveev

Sheet: /USB/USB A Port 2/

File: usb_a_port.kicad_sch

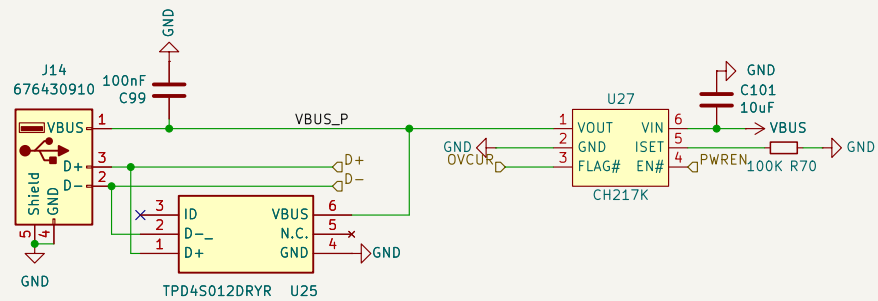
Title: FRANK ZX

Size: A4 Date: 2026-09-29

KiCad E.D.A. 10.0.1

Rev: 1.00

Id: 13/12



<https://github.com/rh1tech/frank>

Mikhail Matveev

Sheet: /USB/USB A Port 3/

File: usb_a_port.kicad_sch

Title: FRANK ZX

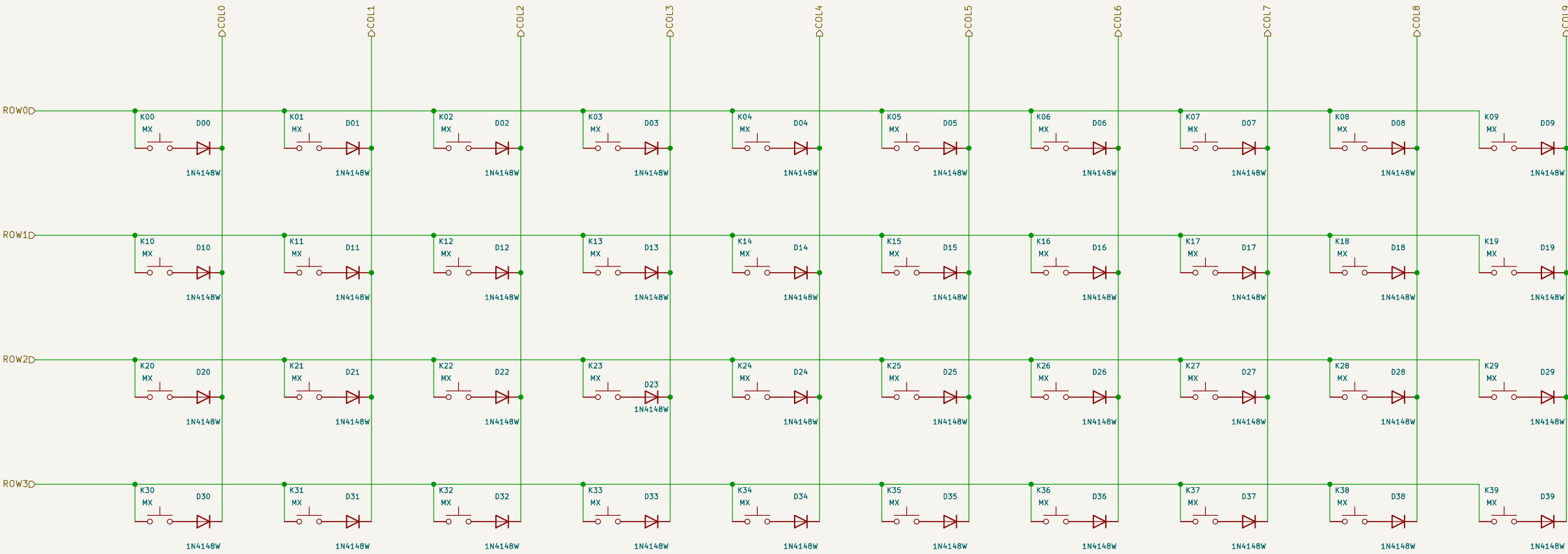
Size: A4

Date: 2026-09-29

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 14/12



40-key matrix (10x4)
Cherry MX + 1N4148W, universal SOD-123 / SOD-323 pads (1N4148WS fits too)
ROW bus above keys (parallel); COL via diodes (cathode to COL)
COL0-9=GPI020-29 ROW0-3=GPI030-33
Scan: drive one COL low, read ROW0-3 with internal pull-ups
(pull-up scheme avoids RP2350 erratum E9 on pull-downs)

<https://github.com/rh1tech/frank>

Mikhail Matveev

Sheet: /Keyboard/

File: keyboard.kicad_sch

Title: FRANK ZX Keyboard

Size: A3

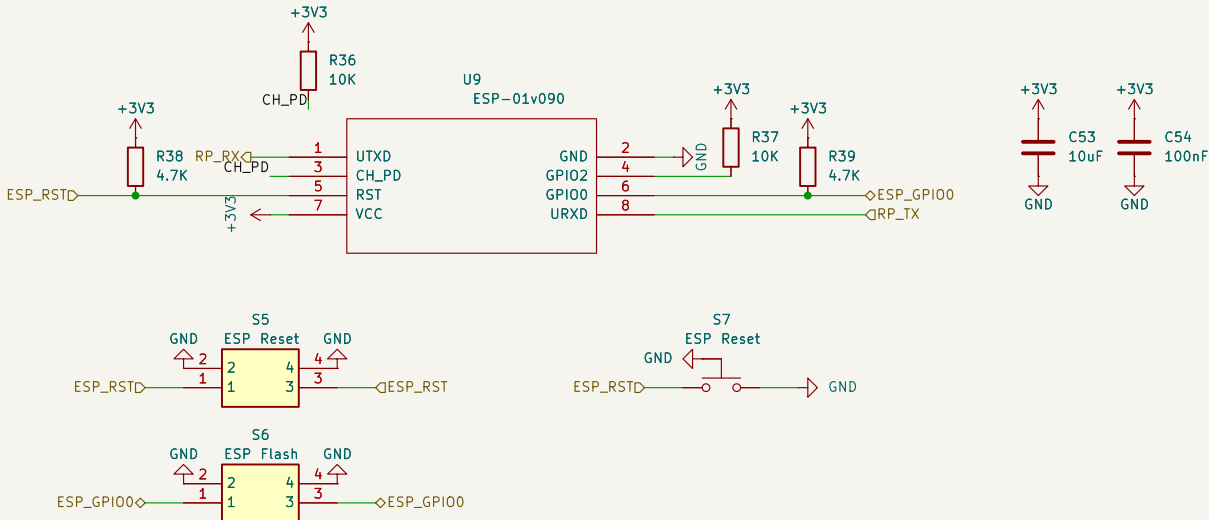
Date: 2026-09-29

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 11/12

ESP-01 (ESP8266) on RP2350B UART1: GPIO36 = UART1 TX -> URXD, GPIO37 = UART1 RX <- UTXD
GPIO38 -> RST and GPIO39 -> GPIO0 (via R12/R13 100R on the RP2350B sheet) let firmware reset and reflash the ESP
R38/R39 4.7K pull-ups win over the RP2350 default pad pull-downs on power-up (normal boot)
C53/C54 at the module VCC pin: ESP8266 TX bursts draw ~300 mA
PCB: keep copper (all layers) out from under the antenna area of the module



S5 resets the ESP-01, S7 is the edge-mounted (TS-1234VW) duplicate of S5
Flash mode: hold S6 (GPIO0 low) while pressing and releasing S5/S7

<https://github.com/rh1tech/frank>

Mikhail Matveev

Sheet: /ESP01/

File: esp01.kicad_sch

Title: FRANK ZX

Size: A4 Date: 2026-09-29

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 12/12